Zero-Logit Model Provenance Auditing via Prober-Generator Resonance Spectroscopy: Privacy-Preserving Pure Cloze Attribution Under Black-Box APIs

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Abstract

To defend against logit-inversion attacks, membership inference, and unauthorized model distillation, modern Large Language Model (LLM) APIs increasingly enforce strict inference privacy by withholding token-level log-probabilities (logits) and internal hidden states. While suppressing logits protects proprietary model weights, it simultaneously blinds white-box forensic detectors (e.g., Fast-DetectGPT, Binoculars) and forces reliance on brittle surface stylometrics or uncalibrated binary thresholds. In this paper, we introduce Prober-Generator Resonance Spectroscopy, a privacy-preserving, zero-logit black-box audit framework that characterizes the macro-discursive signature coherence of generative models purely through text-only multi-sentence cloze infilling ($k \in \{2, 3, 4\}$) and permutation-invariant Kuhn-Munkres bipartite matching. We further demonstrate why applying a fixed binary decision threshold ($\tau = 28.0\%$) with an uncalibrated sigmoidal inflection center ($c_0 = 0.70$) induces domain-dependent calibration drift across specialized human prose, and replace binary thresholding with continuous resonance margins ($\Delta\mu = \mu_{\text{AI}} - \mu_{\text{Human}}$) and threshold-independent separability (AUROC, AUPRC). Evaluated across the $N = 600$ four-slice forensic benchmark (512 AI texts across 6 model families and 88 human academic controls), zero-logit resonance spectroscopy achieves 80.68% pairwise AUROC on DeepSeek/Qwen ($\mu = 45.65\% \pm 17.76\%$), 77.00% pairwise AUROC on Meta LLaMA-3.3 ($\mu = 49.30\% \pm 25.42\%$), 75.69% pooled AUROC (92.99% AUPRC) across the four primary instruction-tuned families ($N = 344$), and 75.76% AUROC (94.91% AUPRC) in Ethical Conundrums, while pooled 6-family AUROC moderates to 68.76% (93.21% AUPRC) due to negative deflection in the high-entropy NVIDIA cohort ($\mu = 19.13\%$ vs. 24.96% human baseline).

1 Introduction: The Inference Privacy vs. Provenance Auditability Dilemma

A central tension in secure machine learning deployment is the conflict between inference privacy and forensic provenance attribution. Exposing token-level log-probabilities (logits) at inference time enables exact vocabulary reconstruction, logit-based model extraction, and training-data membership inference (Carlini et al., 2021). Consequently, frontier LLM deployments restrict API outputs strictly to sampled text tokens. However, state-of-the-art zero-shot detectors such as DetectGPT (Mitchell et al., 2023), Fast-DetectGPT (?), and Binoculars (?) require white-box access to token log-likelihoods or cross-model perplexity ratios. When logits are withheld, practitioners often resort to surface stylometric classifiers (which are easily evaded and penalize formal human prose (?)) or uncalibrated binary score cutoffs.

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1.1 Why Fixed Binary Thresholding Fails Across Specialized Domains

In multi-scale cloze infilling, raw cosine similarity C_{dense} is combined with novel relational predicate overlap P_{novel} via a sigmoidal dynamic gate ($w_{\text{base}} = 0.20, w_{\text{boost}} = 0.40, \kappa = 15.0, c_0 = 0.70$):

$$w_{\text{dense}}(C) = w_{\text{base}} + \frac{w_{\text{boost}} - w_{\text{base}}}{1 + \exp\left(-\kappa(C - c_0)\right)}. \tag{1}$$

When an uncalibrated inflection center ($c_0 = 0.70$) and a single scalar cutoff ($\tau = 28.0\%$) are applied uniformly across heterogeneous domains—such as formal epistemology (Slice 1) or standardized workplace procedures (Slice 4)—domain-shared terminology shifts human baseline representation similarity toward c_0. Consequently, reporting uncalibrated binary Human False Positive Rates (FPR) confounds intrinsic model resonance with domain threshold shift. We therefore formulate Prober-Generator Resonance Spectroscopy, evaluating continuous resonance distributions ($\mu \pm \sigma$), inter-class margins ($\Delta\mu = \mu_{\text{AI}} - \mu_{\text{Human}}$), and threshold-independent AUROC and AUPRC across all $N = 600$ benchmark documents.

2 Methodology: Zero-Logit Pure Cloze Resonance Spectroscopy

Let $D = (s_1, s_2, \ldots, s_L)$ denote an input document partitioned into $L \geq 6$ contiguous sentences. Our engine enforces a strict Zero-Logit, Zero-Stylometric Invariant: no token log-probabilities, punctuation counts, or lexical frequency features are accessed.

2.1 Multi-Scale Contiguous Void Infilling ($k \in \{2, 3, 4\}$)

To isolate macro-discursive logical planning from local token predictability, we excise contiguous multi-sentence spans of length $k \in \{2, 3, 4\}$:

$$B_{t,k} = (s_t, s_{t+1}, \ldots, s_{t+k-1}), \quad D_{\setminus(t,k)} = (s_1, \ldots, s_{t-1}, [\text{VOID_SPAN_k}], s_{t+k}, \ldots, s_L). \tag{2}$$

A black-box reference prober P_{prober} (nvidia/nemotron-3-super-120b-a12b, temperature $T = 0.0$) receives only the text context $D_{\setminus(t,k)}$ and generates a k-sentence infill hypothesis $\hat{B}_{t,k} = (\hat{s}_1, \ldots, \hat{s}_k)$.

2.2 Permutation-Invariant Bipartite Matching & Cross-Span Minimum Gate

Because an infilling model may recover the exact propositional steps of $B_{t,k}$ in a permuted sentence order $\pi \in \mathcal{S}_k$, index-by-index comparison underestimates true structural resonance. We construct the $k \times k$ pairwise similarity matrix $W_{ij} = \Phi(s_{t+i-1}, \hat{s}_j)$ combining dense embedding cosine similarity (all-MiniLM-L6-v2) with novel relational verb-argument overlap P_{novel} (excluding context tokens in $D_{\setminus(t,k)}$), and solve the maximum-weight bipartite assignment via the Kuhn-Munkres (Hungarian) algorithm in $O(k^3)$ time (< 0.1 ms):

$$\mathcal{C}_k(B_{t,k}, \hat{B}_{t,k}) = \max_{\pi \in \mathcal{S}_k} \frac{1}{k} \sum_{i=1}^{k} \Phi\left(s_{t+i-1}, \hat{s}_{\pi(i)}\right). \tag{3}$$

Across evaluated zones $z \in \{1, \ldots, Z\}$ and void horizons $k \in \{2, 3, 4\}$, let p_k denote the scale-specific resonance and $\bar{\mathcal{C}} = \frac{1}{3}(p_2 + p_3 + p_4)$ the multi-scale cloze mean. To penalize localized human creative leaps—where even a single irreconstructible discourse transition collapses the minimum span score—we incorporate the Cross-Span Minimum Gate $S_{\min} = \min_{z,k} \mathcal{C}_k^{(z)}$ into the continuous V3 Resonance score $\mathcal{R}_{\text{V3}}(D) \in [0, 100]$.

3 Empirical Results on the $N = 600$ Four-Slice Forensic Benchmark

We evaluate ClozeEngineV3 on the complete $N = 600$ benchmark (data/eval_v3_n600_forensics_log.jsonl), comprising four thematic slices of $n = 150$ documents each (512 AI texts across 6 model families and 88 human academic controls, 22 per slice).

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Table 1: Continuous Resonance Spectroscopy and Threshold-Independent Separability Across the Four Thematic Benchmark Slices ($N = 600$; 512 AI, 88 Human Controls).

Benchmark Slice	Total (n)	AI / Hu	AI Resonance ($\mu \pm \sigma$)	Human Resonance ($\mu \pm \sigma$)	Margin ($\Delta\mu$)	Slice AUROC	Slice AUPRC
Slice 1: Abstract Foundations	150	128 / 22	36.37% ± 22.67%	24.90% ± 16.90%	+11.48%	64.58%	92.12%
Slice 2: Ethical Conundrums	150	128 / 22	**42.93%** ± **22.89%**	**21.70%** ± **17.27%**	+**21.23%**	**75.76%**	**94.91%**
Slice 3: Strategic Decisions	150	128 / 22	40.69% ± 22.56%	24.31% ± 19.33%	+16.38%	**70.67%**	**93.48%**
Slice 4: Day-to-Day Applied	150	128 / 22	39.32% ± 22.91%	28.93% ± 15.40%	+10.40%	62.93%	91.99%
Combined $N = 600$ Benchmark	600	512 / 88	**39.83%** ± **22.88%**	**24.96%** ± **17.47%**	+**14.87%**	**68.76%**	**93.21%**

Table 2: Multi-Component Signature Decomposition and Pairwise Separability Across All Seven Provenance Classes on the $N = 600$ Benchmark (eval_v3_n600_forensics_log.jsonl).

Provenance Family	N	Scale p_2	Scale p_3	Scale p_4	Cloze Mean ($\bar{\mathcal{C}}$)	Min Span ($S_{\min}$)	Novel Pred (P_{novel})	Fused V3 ($\mu \pm \sigma$)	$\Delta\mu$ vs. Hu	AUROC vs. Hu
Meta (LLaMA 3.3 70B)	84	31.81%	33.13%	33.41%	32.78%	21.91%	22.97%	**49.30%** ± **25.42%**	+**24.34%**	77.00%
DeepSeek / Qwen (V3 / Qwen 2.5)	84	**41.81%**	**44.88%**	**46.41%**	**44.37%**	**27.91%**	**35.41%**	**45.65%** ± **17.76%**	+20.69%	**80.68%**
Google (Gemini 2.5 Flash Lite)	88	31.24%	32.34%	33.33%	32.31%	21.27%	21.60%	**44.35%** ± **24.06%**	+19.39%	72.97%
OpenAI (GPT-4o / 4o-mini)	88	33.49%	34.96%	35.66%	34.70%	22.62%	25.72%	**43.07%** ± **23.62%**	+18.11%	72.41%
Anthropic (Claude 3.5 Sonnet)	84	31.68%	32.57%	33.25%	32.50%	22.97%	21.67%	**37.11%** ± **18.86%**	+12.15%	68.11%
Top-4 Canonical Families Pooled	344	**34.48%**	**36.22%**	**37.10%**	**35.93%**	**23.39%**	**26.36%**	**45.56%** ± **23.04%**	+**20.60%**	**75.69%**
Human Controls (Academic Prose)	88	31.23%	31.90%	32.76%	31.96%	20.91%	20.49%	**24.96%** ± **17.47%**	0.00%	Baseline Floor
NVIDIA (Nemotron 3 Cohort)	84	25.54%	24.90%	25.40%	25.28%	14.69%	13.73%	19.13% ± 9.56%	−5.83%	41.01%

3.1 Key Findings: Discursive Determinism & Bidirectional Deflection

- Ethical Reasoning Maximizes Zero-Logit Separability: In Slice 2: Ethical Conundrums (Table 1), RLHF safety alignment constrains synthetic outputs into highly canonical, multi-step argumentative templates ($\mu_{\text{AI}} = 42.93\%$), whereas human ethical deliberations exhibit sharp philosophical pivots ($\mu_{\text{Human}} = 21.70\%$), yielding a +21.23% separation margin and 75.76% AUROC (94.91% AUPRC).

- High Pairwise Separability on Canonical Families vs. Bidirectional Pooled Moderation: As shown in Table 2, DeepSeek/Qwen achieves 80.68% pairwise AUROC ($\bar{\mathcal{C}} = 44.37\%, S_{\min} = 27.91\%$) and Meta LLaMA-3.3 achieves 77.00% pairwise AUROC ($\mu = 49.30\%$). Across the Top-4 canonical instruction-tuned families ($N = 344$), pooled separability against human prose reaches 75.69% AUROC (92.99% AUPRC). When all six families ($N = 512$) are pooled into a single binary class, AUROC moderates to 68.76% because the high-entropy NVIDIA cohort ($\mu = 19.13\%$) deflects below the human floor (24.96%).

- Refutation of Naive Self-Model Preference: Remarkably, even though our black-box reference prober is nemotron-3-super-120b ($M = 1$), the NVIDIA generative cohort receives the lowest resonance score (19.13%), whereas Meta (49.30%) and DeepSeek/Qwen (45.65%) score more than double. This proves that zero-logit resonance spectroscopy measures the intrinsic discursive determinism $1/H_{\text{discourse}}(p_{\text{gen}})$ of the generator's alignment prior rather than naive architectural self-similarity.

4 Limitations & Tier-2 Inference Privacy Positioning

To maintain strict scientific transparency with zero overstatement, we note three operational boundaries: (1) our empirical evaluation uses a single fixed 120B black-box prober ($M = 1$) across $G = 7$ provenance classes; (2) querying a 120B prober across three void scales requires 2.4–4.0 s per document, positioning our system as an Asynchronous Tier-2 Forensic Audit Protocol rather than a real-time token filter; and (3) under strict zero-logit, zero-stylometric constraints, pooled 6-family separability is moderate (68.76% overall; 75.69% Top-4 families; 80.68% DeepSeek/Qwen), providing interpretable clause-level bipartite attribution evidence rather than black-box binary thresholding.

References

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Table 3: Complete 4×7 Slice $\times$ Provenance Family Resonance Spectroscopy Matrix (Mean V3 Resonance $\mu_{\mathrm{V3}}\%$ across all $N = 600$ samples).

Thematic Slice	Meta	DeepSeek/Qwen	Google	OpenAI	Anthropic	Human	NVIDIA
Slice 1: Abstract Foundations	41.29%	42.55%	38.07%	**45.14**%	31.94%	24.90%	18.69%
Slice 2: Ethical Conundrums	**54.45**%	46.84%	**51.33**%	49.12%	36.74%	21.70%	18.70%
Slice 3: Strategic Decisions	51.49%	44.23%	41.94%	48.42%	39.67%	24.31%	18.18%
Slice 4: Day-to-Day Applied	50.00%	**48.98**%	46.06%	29.58%	**40.09**%	28.93%	20.94%
All-Slice Family Mean (μ)	49.30%	45.65%	44.35%	43.07%	37.11%	24.96%	19.13%

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A Technical Appendix: Mathematical Proofs & Verification

Theorem 1 (Permutation-Invariant Bipartite Optimality). Let $B = (s_1, \ldots, s_k)$ be an excised ground-truth discourse span and $\hat{B} = (\hat{s}_1, \ldots, \hat{s}_k)$ be the prober infill. For any sentence permutation $\sigma \in \mathcal{S}_k$ producing $\hat{B}' = (\hat{s}_{\sigma(1)}, \ldots, \hat{s}_{\sigma(k)})$, the Kuhn-Munkres bipartite operator $\mathcal{C}_k$ satisfies $\mathcal{C}_k(B, \hat{B}') = \mathcal{C}_k(B, \hat{B})$.

Proof. Since $\mathcal{S}_k$ is a finite symmetric group under composition, the mapping $\tau = \sigma \circ \pi$ is a bijection on $\mathcal{S}_k$. Thus $\max_{\pi \in \mathcal{S}_k} \frac{1}{k} \sum_{i=1}^{k} \Phi(s_i, \hat{s}_{\sigma(\pi(i))}) = \max_{\tau \in \mathcal{S}_k} \frac{1}{k} \sum_{i=1}^{k} \Phi(s_i, \hat{s}_{\tau(i)}) = \mathcal{C}_k(B, \hat{B})$. By the Birkhoff-von Neumann Theorem, linear assignment solves this exactly in $O(k^3)$ time. □

Proposition 2 (Domain Calibration Sensitivity of Sigmoidal Gating). If the sigmoidal inflection center $c_0 = 0.70$ in Eq. equation 1 is fixed while evaluating specialized human prose with domain representation shift $\Delta_{\text{domain}} > 0$, the expected human score shifts upward by $\Delta\mathbb{E}[\Phi] > 0$, invalidating fixed binary cutoffs τ. Continuous resonance margins $\Delta\mu$ and threshold-independent AUROC/AUPRC eliminate this artifact.

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